Object Detection Evaluation: From Matching to mAP

Formalized lecture notes for CS 539 · Water-Meter Register Localization

October 2026 | Real-data YOLO evaluation | One-class example: `register`

1 Detection as a Bipartite Matching Problem

For a fixed evaluation dataset, let $G = \{g_1, \dots, g_n\}$ be the set of ground-truth objects, where $g_j = (c_j, B_j)$; let $P = \{p_1, \dots, p_m\}$ be the set of retained detections, where $p_i = (\hat{c}_i, \hat{B}_i, s_i)$ contains a predicted class, bounding box, and confidence score. All objects are indexed by their source image: *matching across different images is forbidden.*

For an IoU threshold $\tau \in [0, 1]$, a candidate edge (p_i, g_j) is admissible only when both objects belong to the same image, $\hat{c}_i = c_j$, and

$$\mathrm{IoU}(\hat{B}_i, B_j) = \frac{|\hat{B}_i \cap B_j|}{|\hat{B}_i \cup B_j|} \geq \tau.$$

The evaluation protocol chooses a *one-to-one* set of admissible matches $M \subseteq P \times G$; each prediction and each GT object participate in at most one match. The specific matching policy (confidence-ordered greedy matching, or another prescribed policy) must remain fixed for comparisons.

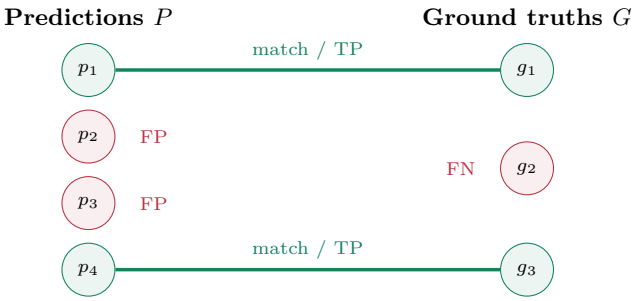


Figure 1: Two views of the *same* matching: TP counts matched *pairs* (green edges); FP counts unmatched predictions; FN counts unmatched GT objects.

$$\boxed{TP = |M|, \qquad FP = |P| - |M|, \qquad FN = |G| - |M|.}$$

A single inaccurate prediction can cause *both* FP (its prediction remains unmatched) and FN (the real object remains unmatched). Conversely, a successful pair contributes *one* TP, not two TPs merely because it can be inspected from two sides.

2 Precision and Recall: Two Denominators, One Matching

$$\boxed{\mathrm{Prec} = \frac{TP}{TP + FP} = \frac{|M|}{|P|}}, \qquad \boxed{\mathrm{Rec} = \frac{TP}{TP + FN} = \frac{|M|}{|G|}}.$$

Precision asks: among *predicted* detections, what fraction is valid? **Recall** asks: among *real* objects, what fraction was recovered? Equivalently, if a GT object is sampled uniformly from G, recall is the empirical probability it has a valid match. When $|P| = 0$, precision requires an evaluation convention; recall is 0 if $|G| > 0$.

Example (the diagram). $|P| = 4$, $|G| = 3$, and $M = \{(p_1, g_1), (p_4, g_3)\}$. Hence $TP = 2$, $FP = 2$, $FN = 1$, $\mathrm{Prec} = 2/4 = 0.5$, and $\mathrm{Rec} = 2/3 \approx 0.667$.

3 Why Introduce a Confidence Threshold?

Each prediction p has score $s(p)$. For cutoff t, retain the *nested prediction subset*

$$\boxed{P_t = \{p \in P_{\text{all}} : s(p) \geq t\}}, \qquad t_1 > t_2 \implies P_{t_1} \subseteq P_{t_2}.$$

At fixed class c and IoU threshold τ, evaluate the retained boxes using the prescribed one-to-one matching $M_{c,t,\tau}$, giving

$$Q_{c,\tau}(t) = \mathrm{Prec}_{c,\tau}(t) = \frac{|M_{c,t,\tau}|}{|P_{c,t}|}, \qquad R_{c,\tau}(t) = \mathrm{Rec}_{c,\tau}(t) = \frac{|M_{c,t,\tau}|}{|G_c|}.$$

Why vary t? A single cutoff hides whether high-confidence detections are actually more reliable than low-confidence ones. Sweeping t measures ranking quality and GT coverage without committing to one arbitrary operating point. t parameterizes the curve but is *not* an axis: $t \mapsto (R_{c,\tau}(t), Q_{c,\tau}(t))$.

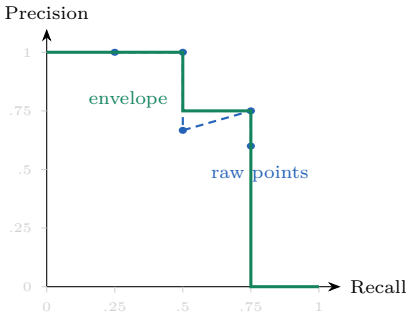
Illustrative ranking comparison. For five identical candidate boxes with matching outcomes $\{TP, TP, TP, FP, FP\}$ and four GT objects, Model A scores the TPs $(.95, .90, .85)$ and FPs $(.30, .20)$; Model B assigns scores $(.20, .30, .40)$ to TPs and $(.90, .95)$ to FPs. Both have overall Prec $= 3/5$ and Rec $= 3/4$ if all boxes are retained, but at $t = .8$, A has Prec $= 1$, Rec $= 3/4$, while B has Prec $= 0$, Rec $= 0$. Ranking matters.

4 Average Precision (AP) at Fixed Class and IoU Threshold

Suppose there are four GT objects and five detections, ordered by descending confidence, whose match outcomes are

$$TP,\ TP,\ FP,\ TP,\ FP.$$

Rank	Result	TP_k	Prec_k	Rec_k
1	TP	1	1	.25
2	TP	2	1	.50
3	FP	2	2/3	.50
4	TP	3	3/4	.75
5	FP	3	3/5	.75



In this example, $Q^*(r) = 1$ on $(0, .5]$, .75 on $(.5, .75]$, and 0 above .75.

As threshold decreases, a new TP raises recall; an FP does not. Precision can fluctuate. Form the *interpolated precision envelope*

$$Q^*_{c,\tau}(r) = \sup_{t:R_{c,\tau}(t)\geq r} Q_{c,\tau}(t),$$

where the envelope is defined as 0 if the required recall cannot be reached. This is a rightward maximum, *not linear interpolation.*
The *continuous-envelope convention* yields

$$\boxed{AP(c,\tau) = \int_0^1 Q^*_{c,\tau}(r)\, dr}, \qquad AP = 0.5(1) + 0.25(.75) + 0.25(0) = \boxed{0.6875}.$$

Specific benchmarks may instead evaluate the envelope on a finite recall grid (e.g., 101 points), resulting in slightly different numeric values. AP summarizes the confidence ranking *and* the attainable recall, rather than averaging raw precision numbers.

5 From AP to mAP: Which Variables Are Averaged?

Keep distinct t (confidence cutoff, swept to form the curve), τ (IoU matching threshold), and c (class). After sweeping t, AP remains a function of c and τ: $AP(c, \tau)$.

$$\boxed{\text{mAP } @\tau = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} AP(c,\tau)} \qquad \text{(mean across classes at fixed IoU threshold).}$$

For the CS 539 register-only detector, $\mathcal{C} = \{\texttt{register}\}$; hence mAP @50 $= AP(\texttt{register}, 0.50)$. Here "50" specifies $\tau = 0.50$ (*not* a confidence threshold or 50% accuracy).
For COCO-style $\mathcal{T} = \{.50, .55, \ldots, .95\}$, the reported summary averages over *both* classes and ten IoU thresholds:

$$\boxed{\text{mAP @50:95} = \frac{1}{|\mathcal{C}|\,|\mathcal{T}|}\sum_{c\in\mathcal{C}}\sum_{\tau\in\mathcal{T}} AP(c,\tau)}.$$

Logical pipeline: candidate boxes + G $\xrightarrow[\tau]{\text{one-to-one matching}}$ $TP/FP/FN$ $\xrightarrow[\text{sweep } t]{P_t}$ PR curve $\xrightarrow{\text{envelope/area}}$ $AP(c,\tau)$ $\xrightarrow{\text{average classes/IoUs}}$ mAP.
Interpretation: mAP@50 rewards approximately correct detection; mAP@50:95 also penalizes box imprecision at stricter localization tolerances. Neither alone measures downstream digit-reading accuracy.